PROJECT PROPOSAL

# PII Filter

Masking personal information before it reaches the LLM

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# Outline


**01** Problem Statement

**02** Background

**03** Motivation

**04** Metric and Benchmark

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# 01

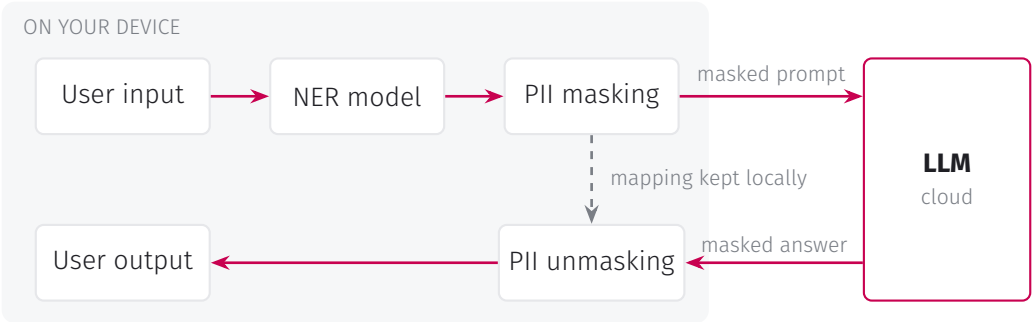
# Problem Statement

# A Concrete Example

| | |
|---|---|
| YOU WRITE | Email Dr. Kim about my March 3 visit. |
| LLM RECEIVES | Email <NAME> about my <DATE> visit. |
| LLM ANSWERS | Dear <NAME>, about my <DATE> visit… |
| YOU READ | Dear Dr. Kim, about my March 3 visit… |

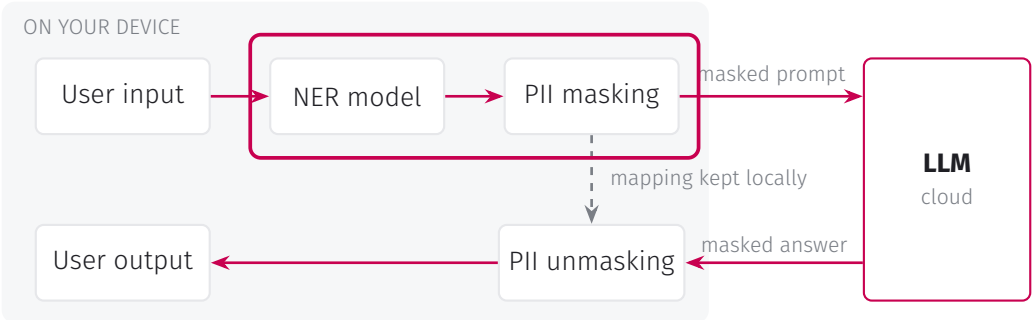
CLOUD

# Proposed Pipeline




The LLM only ever sees placeholders, never the real personal information.

# Proposed Pipeline




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# Problem Statement

## RQ1 Clean text

On clean text, how much better are fine-tuned BERT and GLiNER than rule-based systems across diverse datasets?




## RQ2 Non-standard forms

How much does each system lose when PII is written in a non-standard form?

| | |
|---|---|
| CLEAN | minji@example.com |
| NOISY | minji@exmaple.com |

Rules: Microsoft Presidio (rules only) BERT: gravitee-io/bert-small-pii-detection GLiNER: urchade/gliner_multi_pii-v1

# 02

# Background

# Named Entities and PII

**Named Entity Recognition**

Find the words that name something, and say **what kind of thing** it is.

**PII detection**

The same task, applied to information that can identify a person.

Reach Louis NAME at louis@mail.com EMAIL or 010-1234-5678 PHONE.

Jurafsky & Martin, *Speech and Language Processing*

# From Bi-LSTM-CRF to GLiNER




Each step needs less manual work and adapts better to new kinds of entities.

# PII Masking for LLM Prompts

## Tools

since 2023

Mask the PII, then put it back by simple replacement.

PrivateGPT LangChain LLM Guard

## Research

2023 – 2025

Add a local language model to rewrite or restore the text.

Hide and Seek PAPILLON Rescriber

Private AI 2023 LangChain 2023 Protect AI 2023 Chen et al. 2023 Li et al. 2025 Zhou et al. 2025

# 03

# Motivation

# Who Is Concerned?

## Companies

### Trade secrets

Employees paste code, client data and internal documents into chatbots.

## Hospitals

### Patient records

Medical data is highly sensitive and strictly protected by law.

## Individuals

### Private life

People share names, addresses and health details, often without noticing.

# 04

# Metric and Benchmark

# Benchmarks

## Entities evaluated in this benchmark

| | | |
|---|---|---|
| NAME | EMAIL | PHONE |
| ADDRESS | DATE_OF_BIRTH | CARD_NUMBER |

## Datasets used in this benchmark

**ai4privacy 500k**
Short sentences

**Nemotron-PII**
Forms, emails, notes

**Gretel PII finance**
Financial documents

Ai4Privacy 2025 Steier et al. 2025 Watson et al. 2024

# Metrics

A missed PII leaks. An extra mask only costs some context.

**Precision**

$$P = \frac{TP}{TP + FP}$$

**Recall**

$$R = \frac{TP}{TP + FN}$$

**F1**

$$F1 = \frac{2\,P\,R}{P + R}$$

MAIN SCORE

**F2**

$$F2 = \frac{5\,P\,R}{4\,P + R}$$

# 05

# Experiment Design

# Experiment Design




Every test sentence is scored twice: as it is, and with noise added.

# Toy Experiment

# 06

# Conclusion

# Summary and Next Steps

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[1] Microsoft.
Presidio: Data protection and de-identification SDK.
Python package presidio-analyzer; used as our regex baseline (rule-based recognizers only).

[2] Gravitee.
bert-small-pii-detection.
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BERT-small fine-tuned for PII token classification (25 classes, Presidio label names); used as our BERT model.

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Introducing PrivateGPT, 2023.
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Handling PII data in LangChain, 2023.
PresidioReversibleAnonymizer: anonymize the prompt, de-anonymize the answer with a mapping.

[13] Protect AI.
LLM Guard: Anonymize and deanonymize scanners, 2023.
Placeholders stored in a vault and swapped back in the model output.

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PAPILLON: Privacy preservation from internet-based and local language model ensembles.
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Rescriber: Smaller-LLM-powered user-led data minimization for LLM-based chatbots.
In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2025.
arXiv:2410.11876.

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# Thank you

Questions?